



**Department of Computer Science and Engineering**  
**Final Examination Spring 2023**  
**CSE 321: Operating Systems**

**Duration:** 2 Hours

**Total Marks:** 40

Answer the following questions.  
 Figures in the right margin indicate marks.

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1.</b><br><b>CO4</b> | <p><b>a)</b> In a restaurant, there are 3 washrooms available for male guests and 1 washroom available for female guests. There is a strict rule maintained by the authorities that neither men can use the female washroom nor women can use any of the male washrooms. On a random day during lunch time, 9 male guests needed to use washroom facilities at the very same time. But none of them were allowed to use the female washroom although it was vacant. Therefore, only 3 guests were able to get access to washrooms at a time and others had to wait while maintaining a queue. If any of the washrooms gets vacant, a person from the queue can get access to that. Logically <b>explain</b> which synchronization method has been used here.</p>                                                                                                                                                                                                                                                                                                   | <b>[3]</b> |
|                         | <p><b>b)</b> For Peterson's problem below conditions will be applied.</p> <ul style="list-style-type: none"> <li>• There are two processes: <math>P_1</math> and <math>P_2</math>.</li> <li>• Each Statement takes 4ms to execute, <math>P_1</math> gets executed first</li> <li>• Context Switch will occur after 16ms.</li> <li>• Critical section contains 4 statements.</li> <li>• Remainder section contains 2 statements.</li> <li>• For <math>P_1</math>: <math>i = 0</math> and <math>j = 1</math></li> <li>• For <math>P_2</math>: <math>i = 1</math> and <math>j = 0</math></li> <li>• <math>turn = 0</math></li> <li>• <math>flag[0] = \text{FALSE}</math>, <math>flag[1] = \text{TRUE}</math></li> </ul> <p><b>The structure of process <math>P_i</math> in Peterson's solution:</b></p> <pre style="border: 1px solid black; padding: 10px;">do{     flag[i] = true;     turn = j;     while(flag[j] == true &amp;&amp; turn == 1){         //busy wait     }     //critical section     flag[i] = false;     //remainder section }while(true);</pre> |            |
|                         | <p><b>Complete</b> the table given below for processes <math>P_1</math> and <math>P_2</math> using <b>Peterson's solution</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>[4]</b> |

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
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|                                                                                   | <table><tr><td>Process 1: i = 0, j = 1</td><td>Process 2: i = 1, j = 0</td></tr><tr><td></td><td></td></tr><tr><td colspan="2" style="text-align: center;"></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Process 1: i = 0, j = 1 | Process 2: i = 1, j = 0 |            |  |  |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| Process 1: i = 0, j = 1                                                           | Process 2: i = 1, j = 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| 2.<br>CO4                                                                         | <p>a) Consider the following snapshot of a system:</p> <table><tr><td colspan="2"></td><td colspan="3">Allocation</td><td colspan="2"></td><td colspan="3">MAX</td><td colspan="2"></td><td colspan="3">Available</td></tr><tr><td colspan="2"></td><td>A</td><td>B</td><td>C</td><td colspan="2"></td><td>A</td><td>B</td><td>C</td><td colspan="2"></td><td>A</td><td>B</td><td>C</td></tr><tr><td>P0</td><td></td><td>0</td><td>3</td><td>3</td><td></td><td></td><td>10</td><td>10</td><td>9</td><td></td><td></td><td>8</td><td>5</td><td>6</td></tr><tr><td>P1</td><td></td><td>4</td><td>2</td><td>3</td><td></td><td></td><td>5</td><td>9</td><td>8</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td></td><td>4</td><td>4</td><td>4</td><td></td><td></td><td>9</td><td>6</td><td>7</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td></td><td>5</td><td>4</td><td>4</td><td></td><td></td><td>5</td><td>5</td><td>8</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td></td><td>0</td><td>3</td><td>3</td><td></td><td></td><td>10</td><td>9</td><td>7</td><td></td><td></td><td></td><td></td><td></td></tr></table> |                         |                         | Allocation |  |                                                                                   |     |    | MAX |  |  |           |   | Available |  |  |  |  | A | B | C |  |  | A | B | C |  |  | A | B | C | P0 |  | 0 | 3 | 3 |  |  | 10 | 10 | 9 |  |  | 8 | 5 | 6 | P1 |  | 4 | 2 | 3 |  |  | 5 | 9 | 8 |  |  |  |  |  | P2 |  | 4 | 4 | 4 |  |  | 9 | 6 | 7 |  |  |  |  |  | P3 |  | 5 | 4 | 4 |  |  | 5 | 5 | 8 |  |  |  |  |  | P4 |  | 0 | 3 | 3 |  |  | 10 | 9 | 7 |  |  |  |  |  |  |
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Allocation              |                         |            |  |                                                                                   | MAX |    |     |  |  | Available |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A                       | B                       | C          |  |                                                                                   | A   | B  | C   |  |  | A         | B | C         |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| P0                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                       | 3                       | 3          |  |                                                                                   | 10  | 10 | 9   |  |  | 8         | 5 | 6         |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| P1                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4                       | 2                       | 3          |  |                                                                                   | 5   | 9  | 8   |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| P2                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4                       | 4                       | 4          |  |                                                                                   | 9   | 6  | 7   |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| P3                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5                       | 4                       | 4          |  |                                                                                   | 5   | 5  | 8   |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
| P4                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                       | 3                       | 3          |  |                                                                                   | 10  | 9  | 7   |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   | <p>i. Is the system in a safe state? <b>Apply Banker's safety algorithm</b> to find out the safe sequence. You need to calculate the need matrix.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | [4]                     |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   | <p>ii. <b>P<sub>0</sub></b> requests for <b>(1 4 1)</b>, <b>check</b> the validity of the request. If the request is valid, does the system enter a <b>deadlock</b>?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [1+4]                   |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |
|                                                                                   | <p><b>b)</b> Suppose, in a workplace, we have a set of resource types, <b>R = {R1, R2, R3, R4}</b> and a set of processes, <b>P = {P1, P2, P3, P4, P5}</b>. <b>R1, R2, R3, and R4</b> have <b>2, 3, 1, and 2</b> instances respectively.</p> <ul style="list-style-type: none"><li>• P1 is holding 1 instance of R4 and 1 instance of R1</li><li>• P1 requests 1 instance of R2</li><li>• P2 requests for 1 instance of R4</li><li>• P2 is holding 2 instances of R2 and 1 instance of R3</li><li>• P3 requests for 1 instance of R1</li><li>• P3 is holding 1 instance of R2</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |                         |            |  |                                                                                   |     |    |     |  |  |           |   |           |  |  |  |  |   |   |   |  |  |   |   |   |  |  |   |   |   |    |  |   |   |   |  |  |    |    |   |  |  |   |   |   |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |   |   |   |  |  |  |  |  |    |  |   |   |   |  |  |    |   |   |  |  |  |  |  |  |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                             |         |           |      |           |      |           |     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|------|-----------|------|-----------|-----|
|           | <ul style="list-style-type: none"><li>• P4 is holding 1 instance of R3</li><li>• P4 requests 2 instances of R2</li><li>• P5 is holding 1 instance of R1</li><li>• P5 requests 1 instance of R4</li></ul>                                                                                                                                                                                                                    |         |           |      |           |      |           |     |
|           | <b>Construct</b> a resource allocation graph for the above scenario. <b>Mention</b> the number of cycles found and <b>identify</b> whether there is a deadlock or not.                                                                                                                                                                                                                                                      | [4]     |           |      |           |      |           |     |
| 3.<br>CO5 | a) Given fixed size memory partitions of 300k, 480k, 110k, 200k, 360k, and 550k ( <b>in order, bottom to top</b> ), <b>apply</b> first-fit and best-fit algorithms to place processes with the space requirement of 426k, 300k, 125k, 104k, 475 and 340k (in order). Which algorithm makes the most effective use of memory? Is there any external fragmentation?                                                           | [4+1+1] |           |      |           |      |           |     |
|           | b) Considering in Dynamic memory management technique at a certain time the memory looks like the following figure: <table border="1"><tr><td>OS</td><td>P3 = 200K</td><td>200K</td><td>P6 = 600K</td><td>200K</td><td>P7 = 200K</td></tr></table> Where gray portions of the memory are free spaces. A process P5 = 300K arrives in the ready queue. How can you accommodate P5 in the memory? <b>Justify</b> your answer. | OS      | P3 = 200K | 200K | P6 = 600K | 200K | P7 = 200K | [2] |
| OS        | P3 = 200K                                                                                                                                                                                                                                                                                                                                                                                                                   | 200K    | P6 = 600K | 200K | P7 = 200K |      |           |     |
|           | c) If the page size is <b>3 KB</b> , <b>how many</b> frames will be needed in Main memory for a process size of <b>32,167 Bytes</b> ? Is there any <b>internal fragmentation</b> ? - If yes, <b>calculate</b> the value. [1 KB = 1024 Bytes]                                                                                                                                                                                | [2]     |           |      |           |      |           |     |
| 4.<br>CO5 | a) Suppose there are two processes <b>P1 (16 B)</b> and <b>P2 (12 B)</b> with a page size of <b>4 B</b> . The main memory size is <b>40 B</b> . The page table for P1 and P2 are given below: <div><div>Page table: P1</div><div>Page table: P2</div></div>                                                                                                                                                                 |         |           |      |           |      |           |     |
|           | i. <b>Draw</b> the memory representations consisting of the contents for both P1 and P2.                                                                                                                                                                                                                                                                                                                                    | [2]     |           |      |           |      |           |     |
|           | ii. <b>Find</b> the corresponding physical addresses of the following logical addresses: <div>a) Address 1010 of P1</div> <div>b) Address 0111 of P2</div>                                                                                                                                                                                                                                                                  | [2]     |           |      |           |      |           |     |
|           | iii. <b>How</b> can you make efficient use of the main memory in this scenario?                                                                                                                                                                                                                                                                                                                                             | [1]     |           |      |           |      |           |     |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|  | <p><b>b)</b> Consider a computer with a main memory that has 4 frames and page reference string of 0-7 pages: [2, 5, 7, 5, 2, 0, 2, 5, 6, 2, 6, 6]. The page reference string represents the order in which the pages are accessed by a program. <b>Apply FIFO &amp; OPT</b> algorithm to <b>simulate</b> the page replacement that occurs when the main memory can hold at most 4 pages at a time. <b>Record</b> the number of <b>page faults</b> and compare the result. <b>Mention</b> which algorithm performs better in this scenario.</p> | [4+1] |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|